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Correction to "Post-Quantum Extension of the Conditional CRO Trilemma: Impossibility Bounds on Confidentiality, Reliability, and Opposability Under Quantum Institutional Extraction Limits"

Thierry Emmanuel Minka Mi Nguidjoï[1,2], Flavien Serge Mani Onana[1], and Thomas Djotio Ndié[1]

[1]Laboratory of Mathematical Engineering and Information Systems, National Advanced School of Engineering, University of Yaoundé I, Yaoundé, Cameroon

[2]Supreme State Audit Office, Yaoundé, Cameroon

Corresponding author: Thierry Emmanuel Minka Mi Nguidjoï (minkathierry@gmail.com)

In the above article [1], the proof of Theorem 2 (Post-Quantum Conditional CRO Trilemma) inherits the errors corrected in [3] for the classical Conditional CRO Trilemma [2]: Step 4 uses the classical statement of Lemma 4 of [2], whose inequality has the wrong direction; Step 3 and Assumption 4 use Assumption 2 in the wrong direction; and the constant $C(\beta_q, \alpha_q)$ of Assumption 5 originates from the incorrectly solved optimization step of [2]. In addition, inequality (68) in the proof of Lemma 6 does not hold as written, and Remark 17 overstates the range of validity of Theorem 2. These errors were identified by the authors during the machine-checked formalization of the results in Lean 4. This correction, which parallels [3]: (i) withdraws Lemma 6 and the upper bound (72) of Lemma 7, and restates the lower bound (71) with the Holevo information; (ii) states the verdict-efficiency condition explicitly (Assumption 4′); (iii) states explicitly the leakage regime in which Theorem 2 holds (Assumption 6) and gives a complete proof with explicit constants; and (iv) corrects Remark 17 and clarifies the reading of Section VI-C/D. The quantum institutional model (Assumptions 1–3), the institutional parameters $(\Delta_q, \alpha_q, \beta_q, \eta_q)$, the quantum Fano step (81)–(85), and the form of the main bound (80) are preserved within the stated regime. The authors apologize for any inconvenience caused.

1 Lemmas 6 and 7 (Section V-A)

Lemma 6. Inequality (68), with the expectation taken over *independent* uniform draws of m_0, m_1, does not hold in general: for two messages encoded in orthogonal (classical, perfectly distinguishable) states, $I_{\mathrm{acc}} = 1$ bit, whereas the right-hand side of (68) equals $1/2$. Lemma 6 is therefore withdrawn. The derivation (78)–(79) of the upper bound (72) of Lemma 7 depends on Lemma 6 and is also withdrawn; note that (78) is moreover inconsistent with the linear form (70). Neither Lemma 6 nor (72) is used in the corrected proof below.

Lemma 7. The lower bound (71) follows from the quantum Pinsker inequality and quantum regularity (Assumption 3), with one adjustment: the "simplified form" (74) replaces the Holevo information $\chi(\mathcal{E}_{m,c})$ of Assumption 3 by the accessible information, whereas the Holevo bound gives $I_{\mathrm{acc}} \le \chi(\mathcal{E}_{m,c})$, so that this substitution goes in the wrong direction. Stated with the Holevo information, the lower bound holds and is retained:

Lemma 7 (Quantum confidentiality guarantee; replaces Lemma 7)**.** *For any (β_q, η_q)-quantum-regular protocol $\mathcal{P}$ (Assumption 3),*

$$\mathsf{Conf}^q(\mathcal{P}) \;\ge\; 1 - \frac{1}{2}\sqrt{2\beta_q\,\chi_{\max} + 2\eta_q\,D_{\max}}\,, \tag{71′}$$

where $\chi_{\max} = \max_c \chi(\mathcal{E}_{m,c})$ and $D_{\max} = \max_c D(\bar{\rho}_\Sigma^c \,\|\, \mathbb{I}/d_\Sigma)$; for the NIST PQC schemes of Remark 13, $D_{\max} = \mathsf{negl}(\lambda)$.

Proof. For all m_0, m_1, c, the quantum Pinsker inequality (73) and Assumption 3 give $\|\rho_\Sigma^{m_0,c} - \rho_\Sigma^{m_1,c}\|_1^2 \le 2\beta_q\chi(\mathcal{E}_{m,c}) + 2\eta_q D(\bar{\rho}_\Sigma^c\|\mathbb{I}/d_\Sigma) \le 2\beta_q\chi_{\max} + 2\eta_q D_{\max}$. Taking the maximum over (m_0, m_1, c) in (76) gives (71′). □

This is the quantum counterpart of Lemma 4 as corrected in [3]. Remark 16 is modified accordingly: only the lower bound (71′) is used.

2 Verdict efficiency (Assumption 4 and Step 3)

Assumption 2 gives $\log_2 k_\lambda \le \alpha_q H(C)$, an *upper* bound. Hence neither the last step of (58) nor the second inequality of (90) follows from Assumptions 2 and 4. The lower bound on $H(V \mid C)$ is the efficiency property of institutional verdict functions, which is now stated explicitly.

Assumption 4′ (Verdict non-degeneracy and efficiency; replaces Assumption 4)**.** *The institutional verdict function uses the available context efficiently:*

$$H(V \mid C) \;\ge\; \theta_q\, \alpha_q H(C) \;=:\; \gamma_q, \qquad \gamma_q \ge 1, \tag{57′}$$

for a non-degeneracy constant $\theta_q \in (0, 1]$.

With $\theta_q = 1/2$ and the context entropies of Table 2 ($H(C) \ge 9$ bits, $\alpha_q \ge 0.8$), one has $\gamma_q \ge 3.6$. Claim 1 remains valid through its second argument ($\alpha_q H(C) \ge H(V \mid C) \ge 1$).

3 Corrected statement and proof of Theorem 2 (Section V-B)

The dominant-term analysis of Step 6 assumes $u = x/\lambda = O(\lambda^{-1})$ ("since $x \le C_{\text{proof}} = O(1)$"), while Remark 17 asserts that the bound holds for all $I \in [0, \lambda]$, including $I = \Theta(\lambda)$. The latter assertion is incorrect: for u bounded away from 0, the factor $(\lambda u + \Delta_q)/(\alpha_q H(C))$ in (96) grows linearly with λ. As in [3], the leakage regime is therefore stated explicitly.

Assumption 6 (Confidentiality-preserving regime)**.** *There exists a constant $\kappa \ge 0$, independent of λ, such that*

$$\chi_{\max} = \max_c \chi(\mathcal{E}_{m,c}) \;\le\; \frac{\kappa}{\lambda}. \tag{59′}$$

Since $I(M; \Sigma \mid C)$ is the average of $\chi(\mathcal{E}_{m,c})$ over the contexts and $I_{\text{acc}}(M; \Sigma \mid C) \le I(M; \Sigma \mid C)$ by the Holevo bound, Assumption 6 implies $I_{\text{acc}}(M; \Sigma \mid C) \le I(M; \Sigma \mid C) \le \kappa/\lambda$. By Lemma 7, it is the regime in which quantum-regular protocols guarantee confidentiality: $\mathsf{Conf}^q \ge 1 - \frac{1}{2}\sqrt{2\beta_q \kappa/\lambda + 2\eta_q D_{\max}}$, which tends to 1 when $D_{\max} = \mathsf{negl}(\lambda)$.

Theorem 2 (Post-Quantum Conditional CRO Trilemma, corrected)**.** *Under Assumptions 1–3, Assumption 4′ (57′) and Assumption 6 (59′), for any post-quantum evidence protocol $\mathcal{P}$ with security parameter $\lambda \ge 64$, message entropy $H(M) = \lambda$, M independent of C, and quantum institutional parameters $(\Delta_q, \alpha_q, \beta_q, \eta_q)$:*

$$\mathsf{Conf}^q(\mathcal{P}) \cdot \mathsf{Rel}^q(\mathcal{P}) \cdot \mathsf{Opp}^q(\mathcal{P}) \;\le\; \frac{1}{\gamma_q}\left(\frac{\Delta_q}{\lambda} + \frac{\kappa(\Delta_q + 1)}{\lambda^2} + \frac{\kappa^2}{\lambda^3}\right) \;\le\; \frac{\Delta_q + \eta_q H(C)}{\lambda} + O(\lambda^{-2}) + \mathsf{negl}(\lambda). \tag{80′}$$

Proof. Write $x = I(M; \Sigma \mid C)$; by Assumption 6, $x \le \kappa/\lambda$.

Step 1 (unchanged). By the quantum Fano inequality, (81)–(85) give $I_{\text{acc}}(M; \Sigma \mid C) \ge \mathsf{Rel}^q \cdot \lambda - 1$ for every $\mathsf{Rel}^q \in [0, 1]$. Since $I_{\text{acc}}(M; \Sigma \mid C) \le x$,

$$\mathsf{Rel}^q(\mathcal{P}) \le \frac{x + 1}{\lambda}. \tag{85′}$$

Step 2. By Definition 23, Assumption 1 (87) and Assumption 4′,

$$\mathsf{Opp}^q(\mathcal{P}) = \frac{I(V; \Sigma \mid C)}{H(V \mid C)} \le \frac{x + \Delta_q}{\gamma_q}. \tag{89′}$$

Step 3 (product bound). Since $0 \le \mathsf{Conf}^q(\mathcal{P}) \le 1$ and all factors are nonnegative, $\mathsf{Conf}^q \cdot \mathsf{Rel}^q \cdot \mathsf{Opp}^q \le \frac{x+1}{\lambda} \cdot \frac{x+\Delta_q}{\gamma_q}$. The right-hand side is nondecreasing in x; since $x \le \kappa/\lambda$,

$$\mathsf{Conf}^q \cdot \mathsf{Rel}^q \cdot \mathsf{Opp}^q \le \frac{1}{\gamma_q}\Big(\frac{1}{\lambda} + \frac{\kappa}{\lambda^2}\Big)\Big(\Delta_q + \frac{\kappa}{\lambda}\Big) = \frac{1}{\gamma_q}\Big(\frac{\Delta_q}{\lambda} + \frac{\kappa(\Delta_q + 1)}{\lambda^2} + \frac{\kappa^2}{\lambda^3}\Big).$$

The last inequality in (80′) follows from $\gamma_q \ge 1$ and $\eta_q H(C) \ge 0$. □

Steps 3–7 of the original proof, equations (90)–(111), and Assumption 5 with the constant $C(\beta_q, \alpha_q)$ are replaced by the above; Assumption 5 is no longer needed. The quantum interpretability loss η_q enters the confidentiality guarantee (Lemma 7), the channel bounds (Lemma 10), and the regime classification of Section VI-A, which are unaffected.

Remark (Replaces Remark 17). Assumption 6 is essential. For the classical diagonal encoding of Example 3 (SPHINCS+, $\beta_q = \beta$, $\eta_q = 0$), the protocol that reveals M with probability $1/2$ and otherwise outputs a uniformly random proof satisfies Assumptions 1–3 with $\beta_q \le 1.04$ for all $\lambda \ge 64$ and has $\mathsf{Conf}^q = 1/2$, $\mathsf{Rel}^q = 1/2 + 2^{-\lambda-1}$ and, with V the first bit of Σ, $\mathsf{Opp}^q = 1$, for every λ; here $I(M; \Sigma \mid C) = \lambda/2 - 1 + O(\lambda 2^{-\lambda})$ lies outside regime $(59')$. The bound therefore does not hold uniformly for $I = \Theta(\lambda)$.

4 Clarification of Section VI-C/D and Remark 10

The implied capacities of Tables 8–10, the Dilithium-3 case study (Section VI-D) and item 2 of Remark 10 are obtained by inverting (80) on observed or projected CRO products P_{obs}. Under Assumption 6, Step 1 gives $\mathsf{Rel}^q(\mathcal{P}) \le (1 + \kappa/\lambda)/\lambda$ in the sense of Definition 22, so these products, computed on the operational CLO scale, are not instances of $(80')$. The implied values of Δ_q should therefore be read as an empirical calibration on the CLO scale, consistent with the bounded-rationality range of Section IV-D, and not as consequences of Theorem 2. The operational regimes of Section VI-A, the sensitivity analysis of Section VI-B and the design recommendations of Section VIII are unaffected.

References

[1] T. E. Minka Mi Nguidjoï, F. S. Mani Onana, and T. Djotio Ndié, "Post-quantum extension of the conditional CRO trilemma: Impossibility bounds on confidentiality, reliability, and opposability under quantum institutional extraction limits," *IEEE Access*, vol. 14, pp. 125130–125155, 2026, doi: 10.1109/ACCESS.2026.3722742.

[2] T. E. Minka Mi Nguidjoï, F. S. Mani Onana, and T. Djotio Ndié, "The conditional CRO trilemma: Impossibility bounds on confidentiality, reliability, and opposability under institutional extraction limits," *IEEE Access*, vol. 14, pp. 33263–33274, 2026, doi: 10.1109/ACCESS.2026.3669065.

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